Eros' Metallurgy Lab (EML v4.1): Project Report

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Summary

Eros′ Metallurgy Lab (EML) is a browser-based materials-science and mechanical-design workbench that runs entirely in one web page, with no installation. Version 4.1 combines a hands-on specimen bench, a stress analyzer, six science labs and five engineering-code calculators, and every solver is checked against textbook or measured data each time the page opens.

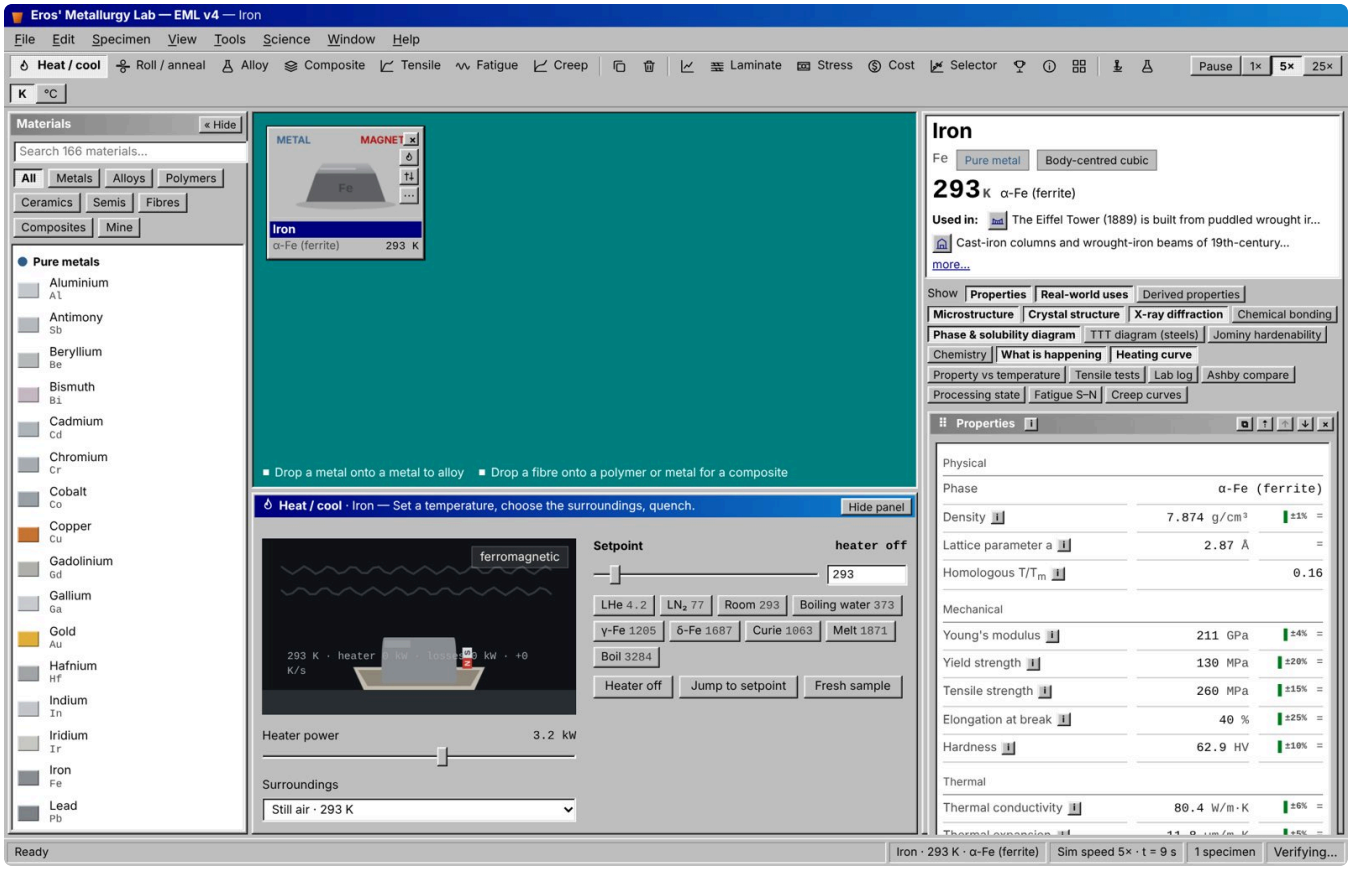
Item	Count
Materials in the database (metals, alloys, polymers, ceramics, fibres, composites)	166
Automated benchmarks (all passing)	80
Science labs (microscope, thermodynamics, kinetics, fracture, corrosion, manufacturing)	6
Engineering codes implemented (EN 1993-1-1, EN 10025-2, VDI 2230, FKM, ISO 286)	5
Encyclopedia entries explaining parameters, methods and limitations	90

It is built for engineering students and young engineers who want to see why materials behave as they do, and to check a design against the methods German industry actually uses.

The workbench

The core of EML is a bench where you drop materials as specimens and process them the way a metallurgist would, while every property updates live.

- **Specimens:** drag any of the 166 materials onto the bench; each card shows its phase, temperature and magnetism.
- **Heat treatment:** heat, hold, quench in water, oil or air, normalise, temper; steels follow their TTT behaviour, and hardness, yield strength and toughness follow the resulting microstructure.
- **Cold work and annealing:** rolling passes raise strength by work hardening, and annealing recrystallises the grains (JMAK kinetics, Hall–Petch grain-size strengthening).
- **Alloying and composites:** drop one metal on another to make an alloy, or a fibre on a polymer to make a composite whose properties come from micromechanics.
- **Virtual testing:** tensile test, fatigue S–N curve (Basquin with surface finish and mean stress) and creep (Norton law, Larson–Miller parameter).
- **Status panels:** properties with source tags and scatter, microstructure, crystal structure, X-ray diffraction, phase diagram, TTT and Jominy curves, bonding, heating curve and real-world uses; each panel can be pinned or popped out as its own window.



Analysis tools

The stress analyzer starts from the shape of the part and then uses the method an engineer would use for that shape.

Tool	What it does	Method
Beams	Rectangle, round, tube, box, I, T and C sections; four support types; point, distributed, axial and torsional loads; colour stress map, shear/moment/deflection diagrams, plastic collapse	Euler–Bernoulli beam elements (120 per beam), exact for these load cases
Plates and brackets	Plate with hole, notches, stepped bar, L-bracket, lug, crack, or a shape you paint; stress concentration factor K_t	2-D plane-stress finite elements (Q4) with stress recovery and mesh convergence
Vessels and rotating parts	Thick cylinders, thin pressure vessels, rotating discs, shafts in torsion	Exact Lamé and disc solutions
Strength-of-materials toolkit	Mohr's circle with Tresca, von Mises, Rankine and Mohr–Coulomb; column buckling; thermal stress; impact loading; shaft design	Euler/Johnson, DE-Goodman
Laminate and sandwich designer	Ply stacking, ABD matrix, first-ply and progressive failure, failure envelopes, sandwich beams	Classical laminate theory with four failure criteria
Material selector	Ashby charts with performance indices and design challenges	Ashby method
Cost estimator	Material plus process cost, including composites	Process cost models

Clicking any point on a stress map sends its stresses to Mohr's circle, and every result shows a safety-factor range that reflects normal material scatter.

Science labs

Version 4 added six labs that compute what textbooks usually only draw, each with its method and data source stated on screen.

Lab	Highlights	Underlying science
Microscope	Etched, polarised, SEM and EBSD views at 100–1000×; pearlite colonies, martensite packets, twins, dendrites, graphite flakes; grain size (ASTM E112) and phase fractions (ASTM E562) measured on the image	Procedural metallography driven by the model state
Thermodynamics	Fe–Cr–Ni ternary sections with tie-lines and lever rule; five binary diagrams; Gibbs-energy curves; why stainless steel is stainless	CALPHAD-style regular-solution model, equilibrium by convex hull
Diffusion and kinetics	TTT and CCT diagrams for real steel compositions with hardness and Jominy curve; carburizing case depth; precipitation ageing of 6061, 7075, 2024 and 17-4 PH	Kirkaldy–Venugopalan model, Fick′s law with Tibbetts diffusivity, Shercliff–Ashby ageing
Fracture mechanics	Remaining life of a crack found in inspection, critical crack size, inspection interval	Paris law with threshold, mean-stress and K_IC effects
Corrosion	Galvanic couples with Evans diagram, pitting risk and time to perforation, stress-corrosion cracking	Mixed-potential theory, PREN, extreme-value statistics, v–K curves
Manufacturing	Casting solidification and shrinkage porosity; welding heat-affected zone and hardness traverse; metal 3-D printing residual stress and warping	Enthalpy finite differences with Niyama criterion, Rosenthal heat source, inherent-strain layer model

Thermodynamics: calculated phase diagrams

Fe–Cr–Ni ternary | Binary diagrams | Gibbs energy curves | Why stainless is stainless

Isothermal section

Temperature 1,100 °C

View: Whole triangle | **Steel corner**

Alloys (click to place)

304 | 316 (Mo as Cr) | 310 | 430 | 410 | 2205 duplex | Invar | Inconel-like

Your composition (wt%)

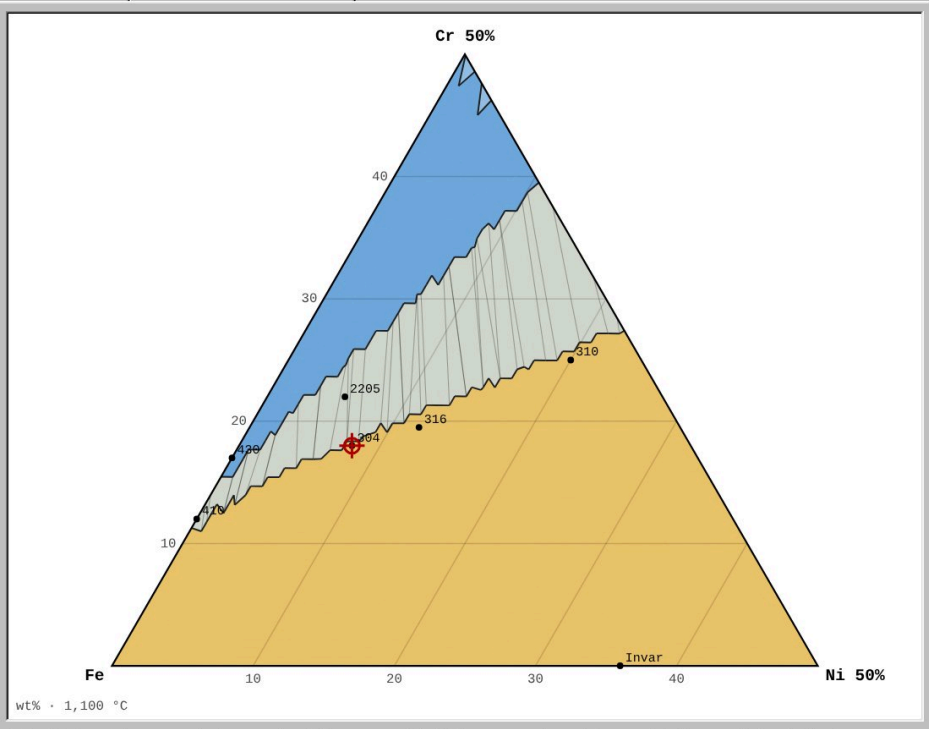
Chromium 18 %

Nickel 8 %

Fe–18Cr–8Ni at 1,100 °C

Phase	Amount	Cr wt%	Ni wt%
austenite γ (FCC)	100%	18	8

Single phase: austenite γ (FCC). Matrix Cr ≥ 10.5%: a continuous Cr_2O_3 passive film can form (stainless).



Calculated, not drawn: each phase gets a Gibbs energy G(x,T) from a regular-solution model; the equilibrium is the lower convex hull of all of them (the common-tangent plane). Tie-lines join the two phase compositions that coexist. Simplified from published CALPHAD assessments of Fe–Cr–Ni (Hillert & Qiu; SGTE unary data). Magnetic terms and the σ phase are left out. Two terms were tuned against known facts: the Fe–Cr FCC term so the γ loop closes near 12 wt% Cr, and a ternary FCC term

Diffusion & kinetics

CCT & TTT diagrams | Carburizing | Precipitation ageing

Steel: AISI 4140 steel

C 0.4 · Mn 0.88 · Si 0.25 · Cr 0.95 · Mo 0.2

Austenite grain size (ASTM G) 7

Austenitizing temperature 850 °C

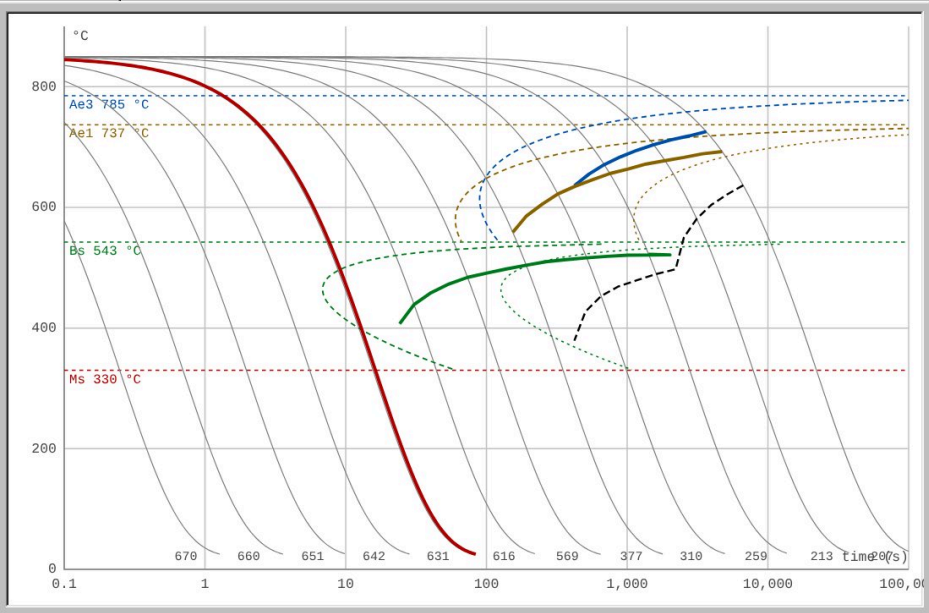
Your cooling curve

$t_{8/5}$ (time from 800 to 500 °C) 8 s

☑ Show isothermal (TTT) curves, dashed

631 HV (57 HRC) | **41.3 °C/s** cooling rate at 700 °C

bainite	0.543%
martensite	96%
retained austenite	3.47%



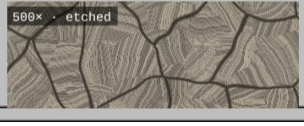
ferrite start | pearlite start | bainite start | 99% transformed | your curve

grey: cooling curves, numbers = HV at the end

Predicted Jominy end-quench curve



Predicted micrograph



Microscope

Microscope | Reading micrographs

Bench specimen | **Classic samples**

Sample: AISI 1080, fully pearlitic

Objective

100× | 200× | 500× | **1000×**

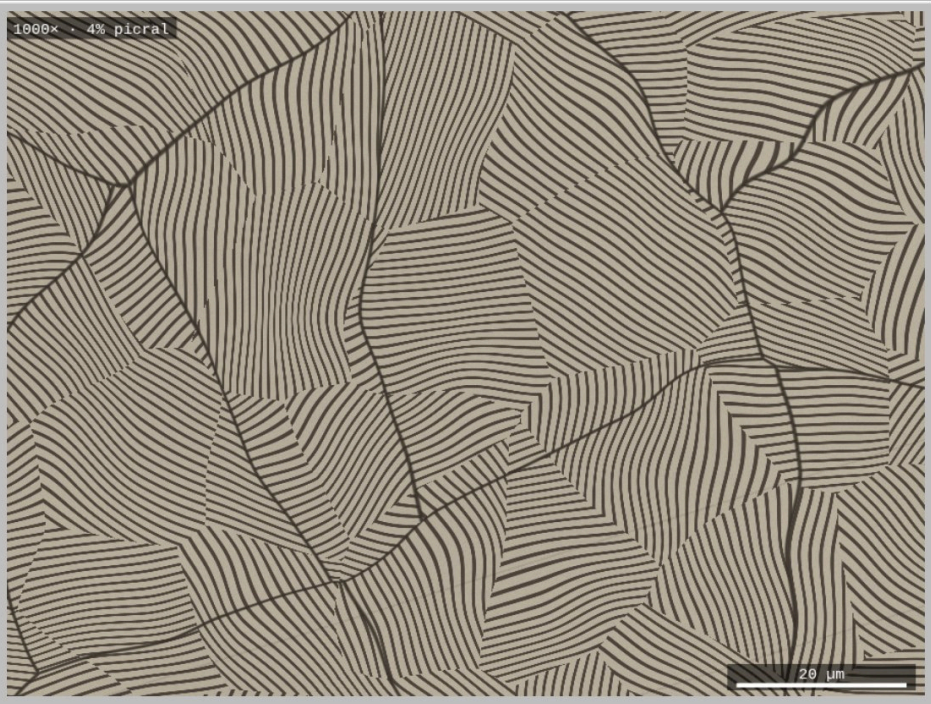
Imaging mode

Etched | Polarised | SEM | EBSD

Measure

Off | Grain size (E112) | Phase fractions (E562)

AISI 1080, fully pearlitic



100% pearlite colonies. At high magnification the alternating ferrite and cementite plates are resolved.

Procedural micrographs generated from the model state (phase fractions, grain size, cold work, cooling rate, arm spacing). Measurements are made on the rendered image exactly as on a real one: the linear-intercept method of ASTM E112 counts grain-boundary crossings on test lines; the point count of ASTM E562 classifies a grid of points. Comparing them with the

Manufacturing simulation

Casting | **Welding** | Metal 3D printing

Base metal: AISI 4140 steel

Process: MIG/MAG (GMAW) · efficiency 0.8

Arc voltage 24 V

Current 220 A

Travel speed 5 mm/s

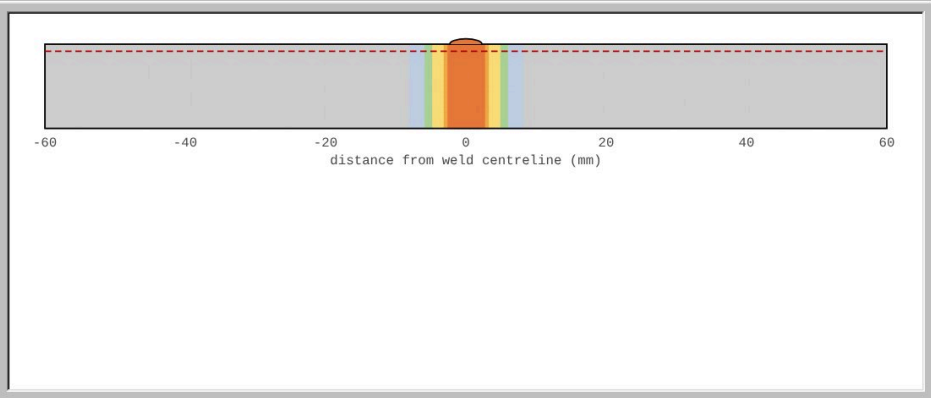
Plate thickness 12 mm

Preheat 20 °C

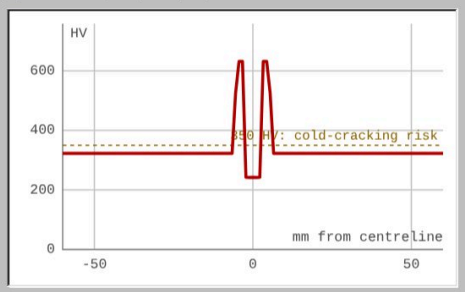
Hardness line: **1 mm below top** | mid-thickness

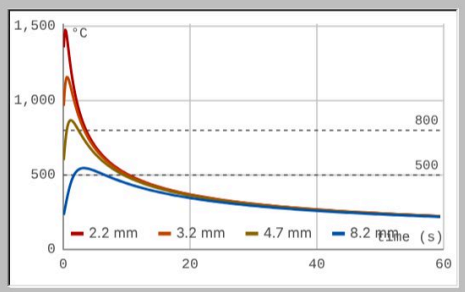
0.845 kJ/mm heat input	**6.72 s** $t_{8/5}$ (thin-plate formula)
4.36 mm fusion-zone width	**3.27 mm** HAZ width (each side)
632 HV max HAZ hardness	**0.777** carbon equivalent CE(IIW)

Hydrogen (cold) cracking risk: HAZ above 350 HV. Preheat (try 299 °C), use low-hydrogen consumables, or raise the heat input to lengthen $t_{8/5}$.



fusion zone | coarse-grained HAZ (>1150 °C) | fine-grained HAZ (Ac3–1150 °C) | intercritical (Ac1–Ac3) | subcritical (tempered) | base metal





▼ **Model validity: 3 ✓ · 1 ⚠**

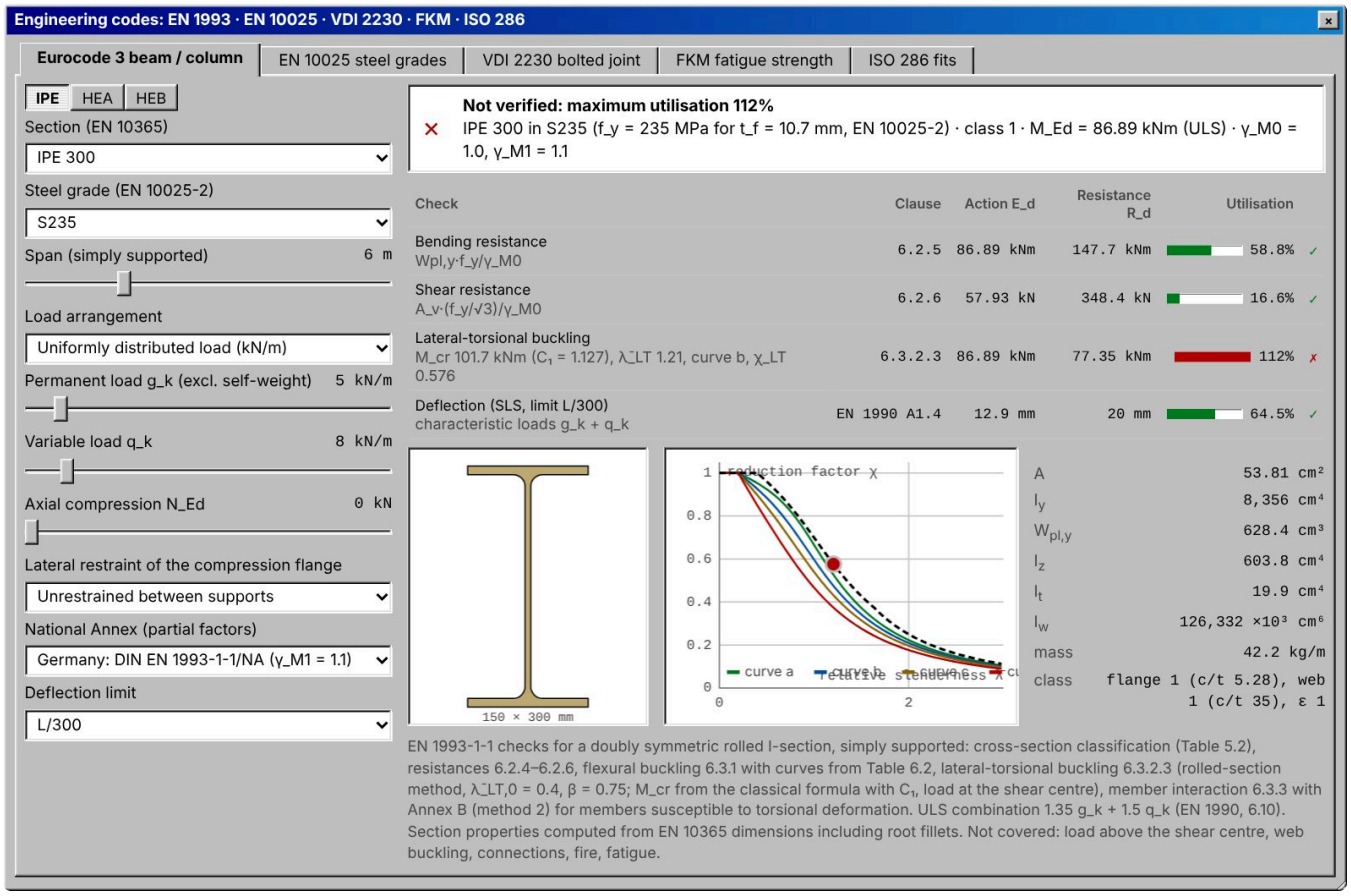
✓ Quasi-steady state (long weld, constant speed).

Engineering codes

EML implements the core equations of five standards used daily in German and European engineering, and presents each result as a clause-referenced check table with a utilisation bar.

Standard	What EML checks
DIN EN 1993-1-1 (Eurocode 3)	IPE/HEA/HEB beams and columns: section class, bending, shear, compression, flexural and lateral-torsional buckling, Annex B interaction, deflection; German National Annex partial factors (γ_M1 = 1.1)
DIN EN 10025-2	S235–S450 yield and tensile strength by thickness, impact grades JR/J0/J2/K2, carbon equivalent
VDI 2230 Part 1	Bolted joints M6–M30, classes 8.8–12.9: tightening torque, load factor, preload, working stress, fatigue, surface pressure, slip (steps R4–R8)
FKM guideline	Fatigue strength of notched machine parts: support number, roughness, surface treatment, mean stress (Haigh diagram), cycles, safety factor; static check
ISO 286-1/-2	Limits and fits up to 500 mm, IT5–IT11, preferred fits H7/g6 to H7/s6, press-fit pressure, torque and shrink temperature

As an example, an unrestrained 6 m IPE 300 in S235 under 5 kN/m permanent and 8 kN/m variable load passes bending at 59% utilisation but fails lateral-torsional buckling at 112%.

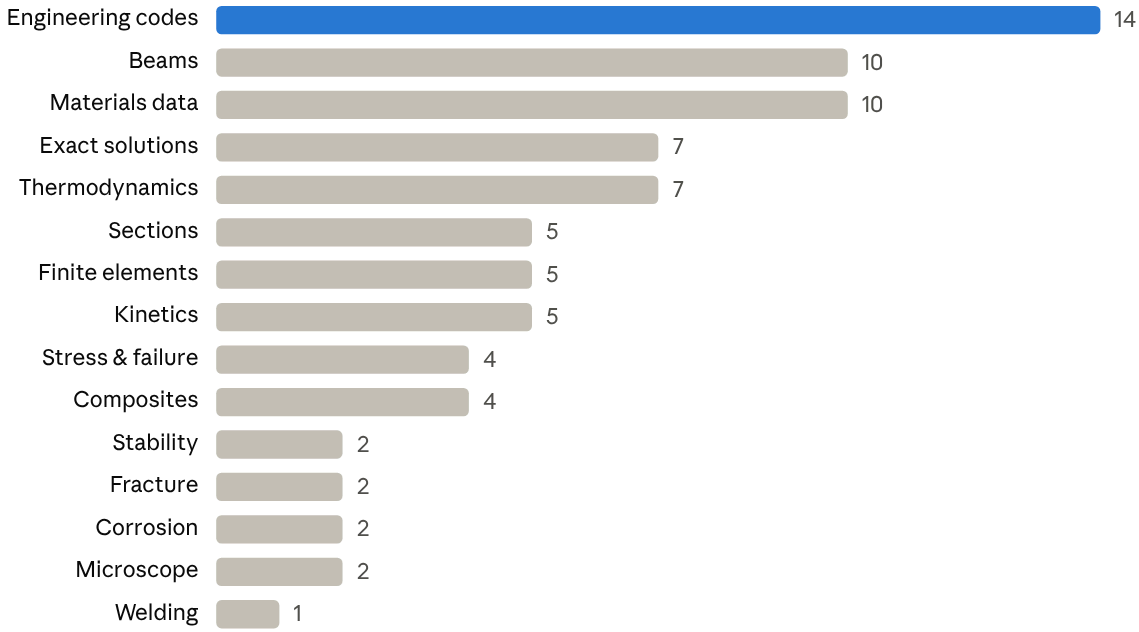


Trustworthiness

Every result in EML can be traced and checked: all 80 benchmarks pass, and each one compares the lab's own solver with an exact textbook solution (verification) or with measured data (validation).

All 80 benchmarks pass across 15 areas

80 of 80 cases within tolerance · run automatically every time the page opens



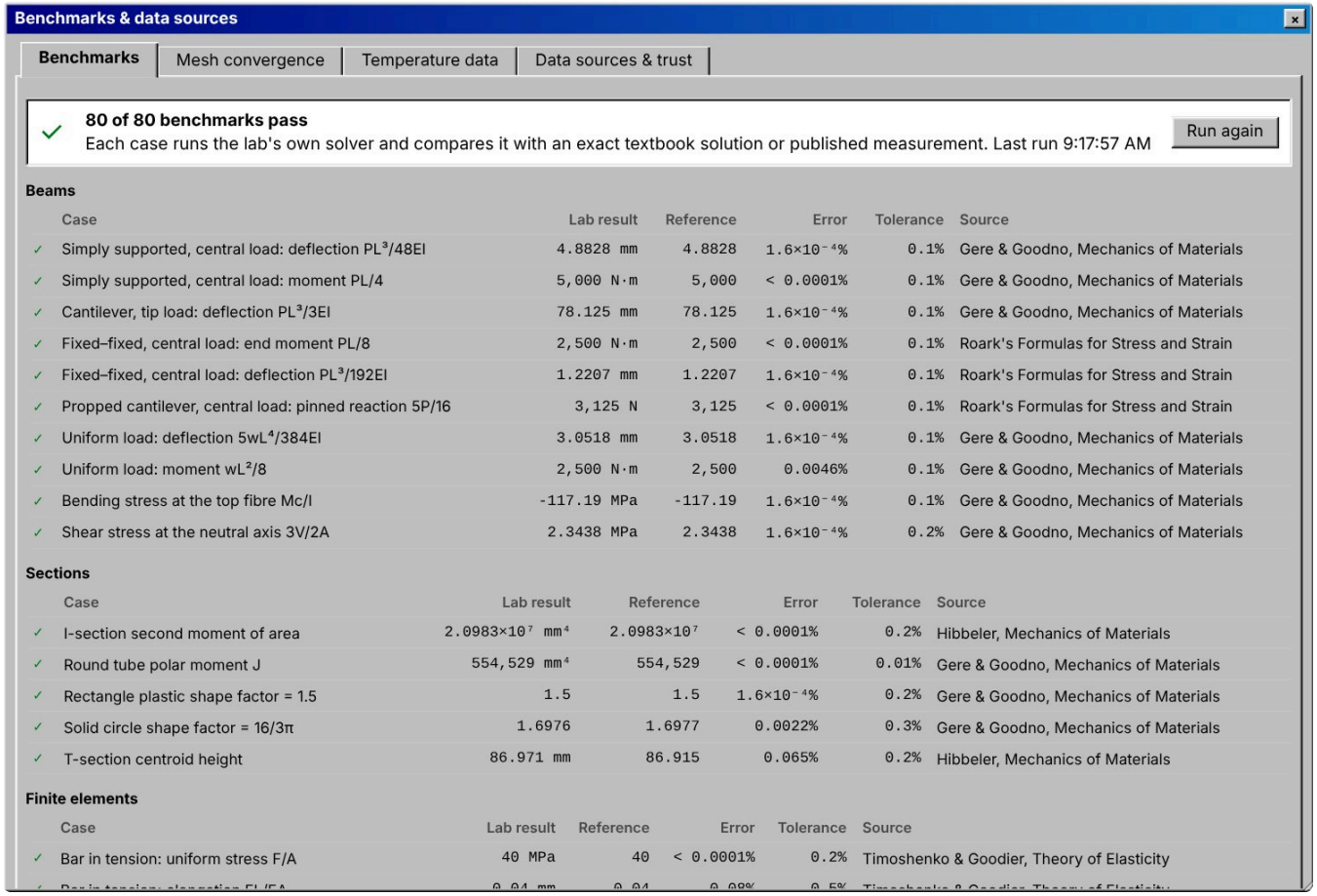
number of benchmark cases per area; the newest area, engineering codes, is highlighted

EML benchmark page, v4.1, run in Chromium on 29 Oct 2026

Validation case	EML result	Reference value	Source
Ag–Cu eutectic temperature	776 °C	779 °C (measured)	Massalski, Binary Alloy Phase Diagrams
Cu–Ni liquidus at 50 wt%	1306 °C	≈ 1312 °C (measured)	Massalski
2205 duplex ferrite at 1050 °C	49%	≈ 50%	Outokumpu handbook
4140 Jominy hardness at 1.5 mm	58 HRC	53–60 HRC band	ASM Handbook Vol. 4
Cu (111) X-ray peak, Cu Kα	43.32°	43.30°	Cullity & Stock
Plate with hole K_t, fine mesh	3.57	3.59	Peterson (Heywood)
IPE 300 second moment of area	8356 cm⁴	8356 cm⁴	EN 10365 section tables
Measured grain size on rendered micrograph	G 7.68	G 7.72 (model input)	ASTM E112

Beyond the benchmarks, three features keep users honest about what the numbers mean:

- **Data provenance:** every property carries a coloured tag (literature value, design-code table, model, estimate, or user-made) and its typical batch-to-batch scatter.
- **Safety-factor ranges:** results show a 5–95% band from material scatter, not just one number.
- **Model-validity checks:** each analysis lists its assumptions and warns when they are broken, for example a coarse mesh, yielding, or a slender beam.



Technology and version history

EML is a single self-contained HTML file of about 980 KB: roughly 6,900 lines of plain JavaScript in about 40 source modules, with all charts, maps, micrographs and solvers drawn on HTML canvas and no external libraries. It runs offline in any modern browser, in light and dark themes, with a classic desktop-style interface of menus, floating windows and pinnable panels.

Version	Added
v4.1	Engineering codes: Eurocode 3 with EN 10025, VDI 2230, FKM guideline, ISO 286; Known Limitations entries in the encyclopedia
v4	Microscope and new micrograph renderer; thermodynamics, kinetics, fracture, corrosion and manufacturing labs; 19 new benchmarks

Version	Added
v3	Benchmarks and data-sources page, data provenance tags, design-code temperature tables, safety-factor ranges, model-validity checks
v2	Shape-first stress analyzer with finite elements, strength-of-materials toolkit, laminate designer, cost estimator, real-world uses, classic desktop theme
v1	Workbench with specimens, heat treatment, alloying, composites, testing, status panels and the parameter encyclopedia

Limitations and future work

EML is a learning and checking tool, not certified design software, and it says so wherever it matters.

- **Property data** are typical handbook values; 27 of the 166 materials use a class estimate for fracture toughness.
- **Calibrated models:** the ageing model is fitted to one handbook peak per alloy, and two Fe–Cr–Ni parameters were tuned to known facts.
- **Indicative results:** stress-corrosion, pitting growth, 3-D printing warping and some Paris-law constants give trends and orders of magnitude only.
- **Code scope:** load above the shear centre (EC3), eccentric loading (VDI 2230), welded parts (FKM) and r/s fits above 120 mm (ISO 286) are not covered.
- **Phones** show a scaled-down desktop layout.

Planned next steps:

1. Printable calculation reports (Rechenbericht) with inputs, equations, clauses and results.
2. German-language interface with standard technical terms.
3. Evaluation of real test data: tensile tests to ISO 6892-1, hardness conversion, Charpy and S–N fits.
4. Monte Carlo reliability analysis and NAFEMS finite-element benchmarks.